

AJAY GOWDA

Phoenix, AZ, USA

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WORK EXPERIENCE

REPUBLIC SERVICES

Phoenix, USA

Sr. Manager, Operations Technology

Sept 2023 – Present

- Build and direct a high-performing, cross-functional team of 6 analysts and managers (spanning IT, Economics, and Automotive backgrounds) to drive enterprise-wide adoption of new fleet technology.
- Directed the rapid development of an AI camera-based safety coaching platform, scaling from prototype to full deployment across the frontline fleet within a year; resulted in an over 80% reduction in safety events.
- Led continuous improvement of billing process and analytics for an AI refuse contamination system processing \$200M+ annually; implemented automated KPI reporting and variance analysis to identify degradation in technology, people, or processes.
- Engineered a semi-automated, scalable data pipeline and billing process for the AI contamination system by aligning Revenue Management and Field Operations, accelerating time-to-value by 6 months and generating \$40M in revenue.
- Evaluate and champion new fleet modernization technologies to ensure continuous operational improvement.

Manager, Technical Operations

Jun 2022 – Aug 2023

- Designed and developed comprehensive EV viability models incorporating route data (miles, lifts), geographic terrain, and custom energy models, directly guiding \$55M+ in capital expenditure (CapEx) across 200+ vehicles and 25 locations.
- Partnered with Oshkosh Corp to engineer the first fully integrated electric refuse vehicle; established customer requirements for improved cab and safety specifications, and successfully piloted the prototype via targeted field training.
- Upskilled non-technical team members in data analytics methodologies, significantly enhancing the team's capacity to deliver streamlined, actionable reporting to cross-functional stakeholders.

EVBOX

Amsterdam, NL

Systems Validation Engineer

Sep 2020 – May 2022

- Architected and deployed an AWS-based centralized data acquisition infrastructure (InfluxDB, Grafana) to ingest and visualize all internal hardware validation test data, enabling rapid analysis and cross-functional decision-making.
- Engineered automated firmware regression testing suites comprising 64 targeted tests, eliminating 2 days of manual execution and report writing per release cycle.
- Designed and built a scalable Hardware-in-the-Loop (HIL) test bench to evaluate charger performance under extreme temperature profiles, delivering critical data that informed product design and significantly reduced production bug rates.

ETERNAL SUN SPIRE

The Hague, NL

Development and Validation Engineer

Aug 2018 – Jul 2020

- Designed and developed control architectures for complex Photovoltaic Test Equipment, ensuring high data accuracy and compliance with AAA testing standards for optimal performance and data acquisition.
- Conducted root-cause analysis and troubleshooting for prototype and production equipment across lab and manufacturing environments, successfully reducing operational downtime by 20%.

GENERAL MOTORS

Milford, USA

Hybrid Powertrain Development Engineer

Jan 2016 – Aug 2018

- Executed comprehensive system-level validation of hybrid and electric powertrains, testing new software releases against rigorous vehicle-level requirements to ensure optimal fuel economy and performance.
- Captured and analyzed millisecond-level telemetry data across all powertrain Electronic Control Units (ECUs) during field tests (test tracks, extreme locations) and advanced test cells to evaluate system behavior.

Algorithm Development Engineer – Thermal Controls

Jul 2015 – Jan 2016

- Developed and rigorously tested thermal control algorithms for liquid-cooled high-voltage battery systems; proactively identified and resolved software anomalies prior to consumer release to ensure product safety and reliability.

EDUCATION

UNIVERSITY OF WASHINGTON

Seattle, USA

Master of Science in ME, Control Systems

Sept 2013 – May 2015

MANIPAL INSTITUTE OF TECHNOLOGY

Manipal, India

Bachelor of Science in Mechatronics Engineering

June 2009 – May 2013

TECHNICAL SKILLS

Languages: Python, MATLAB/ Simulink, JAVA, C/C++, TypeScript, HTML, Structured Text (PLC), YAML, XML, SQL
Technologies: AWS, Grafana, Influxdb, Wireguard, MQTT, REST, Docker, Linux, SCADA, MODBUS, OPC, GIT